

ENTRY 038 Growing the Dataset, Sharpening the Estimate — August 16, 2026 — Status: COMPLETE

Entry 038 — 238 Circuits to 558, and the Sherbrooke/Kyiv Ratio Comes Back Down to Earth

Oversampling Short Routes Triples the Floor-Collapse Sample and Corrects an Overconfident Early Estimate

1. Why Short Routes Specifically

Entry 036 showed floor-collapse doesn't need a long route -- the worst channel-independence gap measured in the whole project came from a 3-edge route. Growing the dataset efficiently therefore means concentrating new circuits in the 1-15 hop range, which is also the cheapest region to simulate (no long SWAP chains).

entry038_grow_shortroute.py reuses Entry 034's exact methodology with a third, independent seed, restricted to four short/mid bins, generating 320 new Bell circuits (160 per chip) on top of Entries 034/035's 238.

2. The Dataset, and the Floor-Collapse Rate, Both Grew

	before (Entries 034/035)	after (+ Entry 038)
total circuits	238	558
floor-collapse cases	12	37
Kyiv short-route rate	3.1%	6.1%
Sherbrooke short-route rate	14.5%	12.5%
Sherbrooke / Kyiv ratio	4.7x	2.05x

The direction of Entry 035's finding holds -- Sherbrooke still collapses on short routes more than twice as often as Kyiv -- but the 4.7x ratio reported on 134 short-route samples was itself noisy. With 396 short-route samples the estimate settles to roughly 2x. This is worth stating plainly: the early number wasn't wrong in direction, but a batch of ~130 samples was too small to trust its exact magnitude, which is exactly the kind of correction a bigger dataset is supposed to produce.

3. Re-Running Entry 037's ML Baseline on the Larger Sample

model	MAE, all circuits	MAE, floor-collapse only	n floor-collapse
v4.1 (closed form)	4.60 pts	18.20 pts	37
Gradient boosting (CV)	4.57 pts	14.79 pts	37
Random forest (CV)	4.36 pts	15.34 pts	37

With 37 floor-collapse examples instead of 12, every cross-validation fold now contains floor-collapse cases (5 of 5, versus an uneven 2-3 before) -- a materially more trustworthy estimate. The relative gain holds: gradient boosting cuts floor-collapse MAE by about 19%, random forest by about 16%, consistent with Entry 037's ~22% finding on the smaller sample. Overall MAE for every model, v4.1 included, dropped as well (6.3 to ~4.5 points) -- this batch's short-route bias makes the whole dataset easier on average, not because anything got fixed.

Entry 038 — Conclusion

Growing the dataset did what it was supposed to: it didn't just add volume, it corrected an overconfident early ratio (4.7x to 2.05x) and confirmed the ML baseline's floor-collapse advantage is real and consistent, not an artifact of a 12-example sample. The floor-collapse phenomenon itself remains exactly as unexplained as Entry 036 left it -- this entry is about data quality and confidence, not a new mechanism.

4. Next Step

Continue growing toward the low thousands using the same resumable scripts (Entries 034, 035, 038 all checkpoint to JSON and can resume in chunks), then retrain and re-evaluate the ML baseline again -- watching whether the floor-collapse MAE gap keeps narrowing or plateaus, which would itself be informative about whether a tabular model

can ever fully close it or whether it genuinely needs correlated-noise modeling as Entry 036 suggested.

Research Log — Index Addendum (Entry 038)

Entry	Title	Date	Status
Entry 038	Growing the Dataset to 558 Circuits, Refining the Floor-Collapse Rate	Aug 16, 2026	Complete

Script: entry038_grow_shortroute.py. Data: quantumbridge_data/entry038_shortroute_dataset.json (320 new circuits), quantumbridge_data/entry038_combined_dataset.json (558 total, merged), quantumbridge_data/entry038_growth_summary.json. Sherbrooke/Kyiv floor-collapse ratio revised from 4.7x to a more reliable 2.05x; ML baseline's floor-collapse advantage confirmed with full 5-fold coverage.